R-SESSIONS

FEATURE OVERVIEW AND USER GUIDE

PRODUCT R-Sessions: SSH, SFTP, Telnet, Serial and VNC session manager

VERSION 1.0.0 and later

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PREPARED BY RALKOM Rafał Lesiński

RALKOM Rafał Lesiński • ul. Święty Marcin 28/9, 62-806 Poznań, Poland
VAT ID PL7811664559 • tel. +48 572 210 250 • kontakt@ralkom.pl • www.ralkom.pl

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Part I. Program overview

How to use this document

Part	For whom	Contents
I. Program overview	business owners and purchase decision makers	what the program does, how it protects data, how the Free and Pro editions differ
II. User guide	people who work with the program every day	first start, the main window, hosts, sessions, files, tunnels, snippets, Telnet, serial port, VNC
III. Administration	system administrators and IT support	installation, SSH keys and agent, jump hosts, import and export, settings, license, security, troubleshooting

Names of buttons, fields and commands are given exactly as they appear in the program, in bold, e.g. **New host**. Terms you do not need to know for everyday work are explained in "What is…?" boxes. All data shown in the figures (the company Nordhaven Logistics, its servers and addresses) is fictional.

1. What is R-Sessions

R-Sessions is a program by RALKOM for working with servers and network devices. In a single window, on tabs, it opens SSH terminals, files on servers (SFTP), Telnet connections, device consoles over a serial port and the desktops of remote computers (VNC). It works the same way on Windows and Linux.

Your server list, passwords, SSH keys, saved commands and tunnels are kept in a **vault**: an encrypted file on your computer, opened with a password. The program needs no account and no cloud connection, and it never sends data about you or your servers anywhere.

The program is designed for administrators and IT teams, but it also suits people who connect to just a few servers and do not know the networking details: save a host once, then connect to it with a double-click.

2. What it supports

Connection type	What it is for	Notes
SSH	a terminal (command line) on a Linux or Unix server, a router or a switch	encrypted connection; sign-in with a password, a key or an SSH agent; jump hosts, tunnels, server statistics
SFTP	files on an SSH server in a two-pane window	transfer queue, resuming interrupted transfers, file editing, Root mode (sudo)
Telnet	older network devices and console servers	unencrypted connection, clearly marked in the program

Connection type	What it is for	Notes
Serial	a device console over a serial port (COM, USB adapter)	speed, data bits, parity, flow control, Break signal
VNC	the graphical desktop of a remote computer	VNC password; a connection through an SSH server is encrypted
Local shell	the command line of this computer (e.g. PowerShell, bash) on a tab next to remote sessions	no network connection

3. Security and privacy

- **Vault with a mandatory password.** Hosts, passwords, private keys, snippets, tunnels and settings are encrypted on disk. Without the vault password nobody, RALKOM included, can read this data.
- **Proven encryption.** The password is turned into a key with the Argon2id algorithm, and data is encrypted with XChaCha20-Poly1305. Every saved item (host, password, key) has its own encryption key.
- **Recovery code.** When you create the vault, the program gives you a code that opens the vault if you forget your password. It is the only way to recover your data.
- **Automatic lock.** The vault locks after a period of inactivity and when the computer is locked or goes to sleep.
- **Server verification.** The program remembers the key fingerprint of every SSH server and warns you when the key changes, which may indicate an eavesdropping attempt.
- **No account, no cloud, no telemetry.** The license is a file checked on your computer, with no online activation. The program sends no usage statistics and no information about your servers.
- **Digital signature.** Windows installers are signed by RALKOM, and every update installed from within the program is checked against the update signing key before installation, on both Windows and Linux.
- **Logs without passwords.** On request the program records sessions to a file, but only what the server printed, never the characters you type, so passwords you enter never end up in logs.

4. Everyday convenience

- Session tabs, split panes and moving tabs to separate windows.
- Search with `Ctrl` + `K`: a host, command, tunnel or snippet in a few keystrokes.
- **Quick connect** to a server that is not on your list, without saving it.
- Snippets: saved commands with variables, run in one session or in several at once, also through sudo.
- SSH tunnels with a clear diagram, started manually, together with the host or always on.
- Server statistics bar: CPU, memory, network, disk and uptime.
- Two-pane file manager with a transfer queue, resuming and checksum verification of large files.
- Host import from OpenSSH and PuTTY, and moving your whole configuration to another computer in an encrypted file.

- Connection history, a bell with all notifications, Dark, Graphite and Light themes or one that follows the system.
- Automatic updates that never close your sessions without asking.

5. Free and Pro editions

R-Sessions comes in two editions. The Free edition costs nothing and includes every feature; it only limits the number of saved hosts and tunnels. The Pro edition is enabled by a license file.

	Free	Pro
Saved hosts	up to 15 (local shells do not count)	unlimited
Saved tunnels	up to 2	unlimited
Quick connect	unlimited	unlimited
All other features (SSH, SFTP with sudo, Telnet, Serial, VNC, split panes, snippets, logs, import and export, keys, vault, lock)	yes	yes
Fixes and security updates	yes	yes
New Pro features released after version 1.0	no	released up to the license's "New features until" date

- The license does not expire. The **New features until** date only says until when the license covers new Pro features. After that date the program keeps working in full, with current fixes.
- The license is issued for a number of seats. **One seat is one person**, who may use the program on any number of their own computers.
- The program never deletes or locks data because of the license. If you have more hosts than the limit (e.g. after removing a license), everything keeps working; you just cannot add new ones.

Part II. User guide

6. First start

> **What is the vault and the recovery code?** The vault is an encrypted place on your computer where R-Sessions keeps all its data: your server list, passwords, keys and settings. It is opened with the vault password, which you enter every time you start the program. The recovery code is a second, long key to the same vault, in case you forget the password. Nobody but you knows the password or the code.

Creating the vault

On first start the program asks you to create a vault.

1. In the **Create your vault** window, type a password in **Password** and repeat it in **Repeat password**. The password must be at least 10 characters long. The bar below the field shows its strength.
2. Click **Create vault**.
3. The program shows the **Your recovery code** window. Save the code with **Save as file…** or copy it with **Copy**, and keep it away from this computer, e.g. printed and stored in the company safe, or in a password manager.
4. Select **I have saved my recovery code** and click **Continue**.



Figure 1. The Create your vault window: vault password and strength meter (sample data)

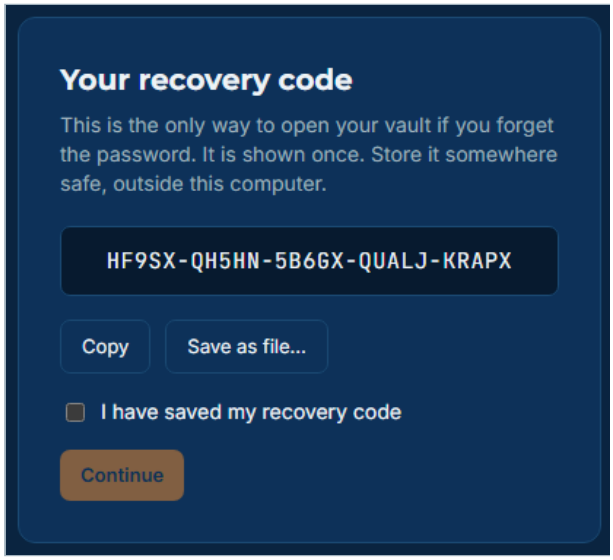


Figure 2. The recovery code, shown only once, after the vault is created (sample data)

> **Note.** The recovery code is shown only once. Without the password and the code, the vault cannot be opened and the data in it cannot be recovered, not even with RALKOM's help. You can generate a new code at any time in **Settings** → **Security** (see "Settings").

Locking and unlocking

A locked vault hides your data behind the **Unlock R-Sessions** screen. Open connections are not interrupted: after unlocking, your sessions are where you left them. The vault locks:

- manually, with the padlock button in the top right corner of the window (**Lock vault**) or with `Ctrl` + `Shift` + `L`;
- after the period of inactivity set in **Settings** → **Security** → **Auto-lock** (Off, 5, 15, 30 or 60 minutes);
- when the computer is locked or goes to sleep, if **Lock when the computer locks or sleeps** is enabled (it is by default).

To unlock the vault, type your password in **Password** and click **Unlock**.

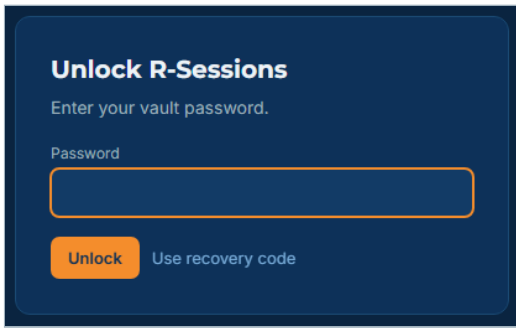


Figure 3. The unlock screen with the Use recovery code button (sample data)

Forgotten password

1. On the unlock screen, click **Use recovery code**.
2. Type the recovery code in **Recovery code** and click **Unlock**.
3. The program immediately asks for a new password (the **Set a new password** window). Type it twice and click **Set password**.
4. The program shows a new recovery code. Save it the same way as when you created the vault. The code you used stops working.

7. The main window

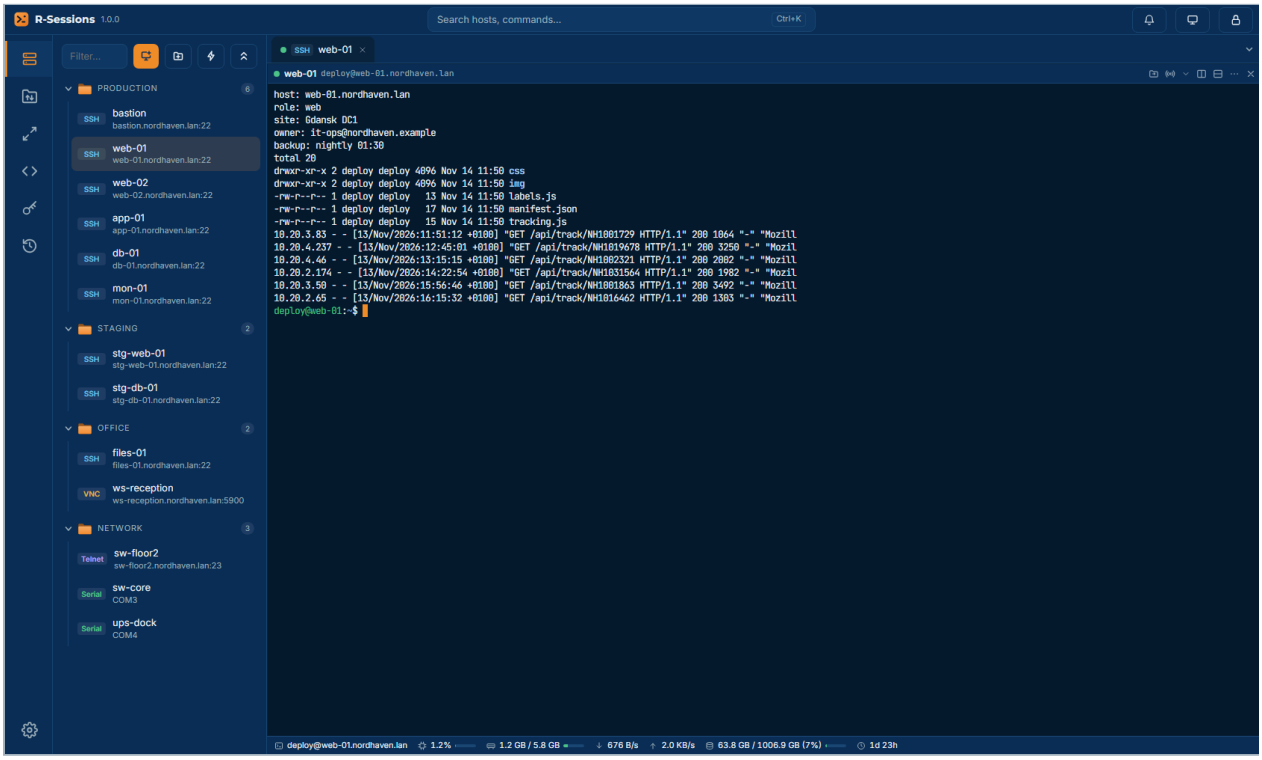


Figure 4. The main window: host groups on the left, an open SSH session with the web-01 server and the statistics bar at the bottom (sample data)

Element	What it is for
Title bar	program name and version, search (**Search hosts, commands…**), the notification bell, the theme button and the vault lock padlock
Icon bar (left)	switches sections: **Hosts**, **SFTP**, **Tunnels**, **Snippets**, **Keychain**, **History**, and at the bottom **Settings** (gear)
List panel	contents of the selected section, e.g. hosts in groups, with a **Filter…** field and buttons at the top
Tabs	open sessions; the dot next to the name shows the state (green: connected, red: connection lost)
Pane bar	session name and address, plus buttons: open SFTP, broadcast, split, more actions, close pane
Statistics bar	session address and server data: CPU, memory, network, disk, uptime (see "SSH sessions and the terminal")

The theme button (next to the bell) cycles through the **Dark**, **Graphite**, **Light** and **System** (follows the system setting) themes. Once a Pro license is loaded, the program name in the title bar and in the window title changes to "R-Sessions Pro".



Figure 5. The title bar with a Pro license (sample data)

Ctrl+K search

Click the **Search hosts, commands…** field or press `Ctrl` + `K` and start typing. The list shows matching hosts (`Enter` connects), recently used hosts (**Recent**), commands (**New host**, **New group**, **Open local shell…**, **Open SFTP…**, **Quick connect…**, **Lock vault**, **Settings**, **Quit R-Sessions**), tunnels (**Start tunnel: …**, **Stop tunnel: …**, **New tunnel**) and snippets (**Run snippet: …**). When the text you type looks like an address, e.g. `admin@10.20.0.5`, a **Connect to …** item appears.

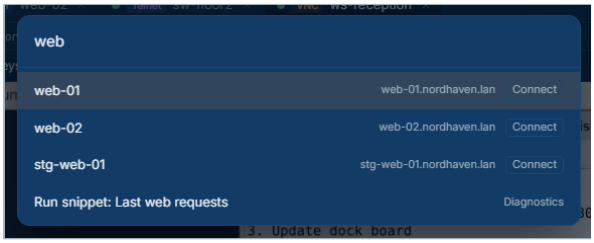


Figure 6. Search after typing "web": three hosts and a snippet (sample data)

> **Tip.** In the terminal, `Ctrl` + `K` belongs to the server's shell. There, `Ctrl` + `Shift` + `K` opens search, and `Ctrl` + `Shift` + `P` opens a list of snippets only.

Tabs and windows

Each session opens on a new tab. Close a tab with its X, the middle mouse button or the **Close tab** command. Tabs can be dragged, and when there are many, the arrow button at the right end of the tab bar (**All tabs**) shows a searchable list of all tabs. The tab context menu (right mouse button) contains:

Command	Action
Snippets	runs a saved snippet in this session
Type password	types one of the user passwords saved with the host into the session (see "Hosts and groups")
Send Break	serial port only: sends a Break signal
Save as host	Quick connect only: saves the connection as a host
Move to new window	moves the tab to a separate window (e.g. on a second monitor); **Move back to main window** returns it
Close tab	closes the tab

Keyboard shortcuts

Shortcut	Action	Shortcut	Action
`Ctrl` + `K`, `Ctrl` + `Shift` + `K`	search	`Ctrl` + `Shift` + `O`	SFTP files of the current session's server
`Ctrl` + `Shift` + `P`	run a snippet	`Ctrl` + `Shift` + `C`, `Ctrl` + `Shift` + `V`	copy and paste in the terminal
`Ctrl` + `Shift` + `N`	Quick connect	`Ctrl` + `Shift` + `F`	find in the terminal
`Ctrl` + `N`	new host (in the host list)	`Alt` + `Shift` + `=`, `Alt` + `Shift` + `-`	split right, split down
`Ctrl` + `Shift` + `L`	lock the vault	`Ctrl` + `Shift` + `W`	close the pane

File shortcuts (`F5`, `F6` and so on) are described in "Files (SFTP)".

8. Hosts and groups

> **What is a host?** A host is a saved connection description: the name shown in the list, the address of the server or device, the connection type (e.g. SSH) and the sign-in method. With it you connect with a double-click, without typing the address and password.

Adding a host

1. In the **Hosts** section, click the orange **New host** button above the list or press `Ctrl`+`N`. The **New host here** command in a group's menu adds the host directly to that group.
2. Enter a **Label** (the name in the list), choose the **Protocol**, and enter the **Address** (name or IP address) and **Port**.
3. In **Identity**, choose a saved identity with a username and sign-in method, or **+ Create new…**. Without an identity, the program asks for the username and password when connecting.
4. Choose a **Group** and click **Save**.

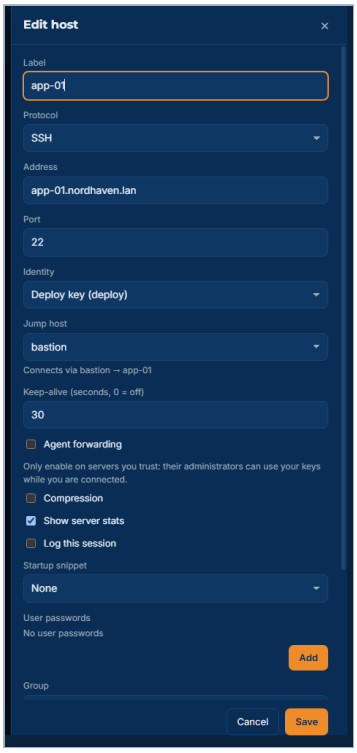


Figure 7. Editing the SSH host app-01: Deploy key identity and the bastion jump host (sample data)

SSH host field	Meaning
Identity	username and sign-in method (password, key, SSH agent); one identity can serve many hosts
Jump host	an intermediate server (bastion) the connection goes through; the full route is shown below the field, e.g. "Connects via bastion → app-01" (see "Jump hosts")
Keep-alive (seconds, 0 = off)	how often to send a keep-alive signal so a firewall does not drop an idle connection
Agent forwarding	makes your SSH keys available to the server for the session (see "Keychain"); enable only on trusted servers
Compression	data compression, useful on a slow link
Show server stats	the server statistics bar below the terminal
Log this session	records the session to a file (see "Session logs")
Startup snippet	a snippet run automatically after connecting
User passwords	extra passwords for this host, e.g. for the root account or a device account, typed with the **Type password** command; **Add** button (available once the host is saved)
Group, tags, **Notes**	place in the list, labels for search, and free-form notes

A user password with **Use for sudo** selected is also typed by the program itself when a snippet run through sudo asks for a password (see "Snippets").

Host list and groups

Open a host with a double-click or `Enter`. `F2` opens the editor, and `Delete` deletes the host after confirmation. The **Filter…** field above the list searches names, addresses and tags. The buttons above the list are, in order: **New host**, **New group**, **Quick connect** (lightning bolt) and collapse or expand all groups.

A group has a name, a color and a **Parent group**, so groups can be nested. Move a host to another group by dragging it or with **Move to group…**. The group menu contains **New host here**, **New subgroup**, **Edit** and **Delete**; when you delete a group, the program asks where to move its contents (**Move contents to**).

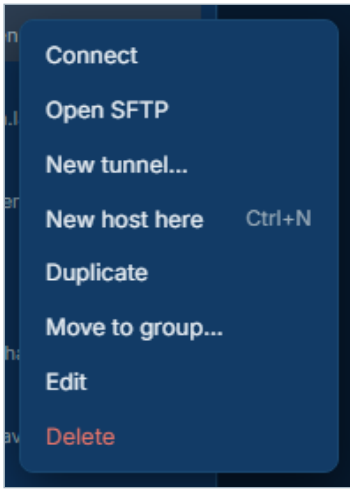


Figure 8. The SSH host context menu (sample data)

The host menu (right mouse button) contains: **Connect**, **Open SFTP** and **New tunnel…** (SSH only), **New host here**, **Duplicate** (a copy of the host to adjust), **Move to group…**, **Edit** and **Delete**.

Quick connect: connecting without saving a host

Quick connect opens an SSH session with a server that is not on your list. It works without limits in both Free and Pro.

1. Click the lightning bolt button above the host list or press `Ctrl` + `Shift` + `N`.
2. In **Address**, type the address as `user@host:port`; user and port are optional (the default port is 22).
3. Optionally choose an **Identity** and click **Connect**. Without an identity, the program tries the SSH agent and then asks for a password, which it does not save.

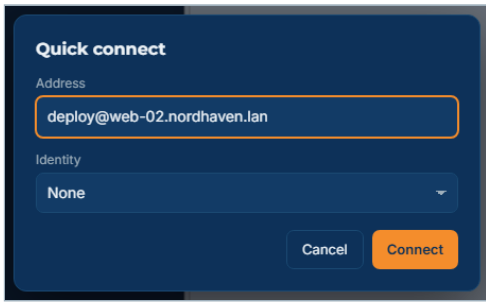


Figure 9. The Quick connect window (sample data)

The connection is added to the history. To keep it permanently, choose **Save as host** from the tab menu. Quick connect does not support jump hosts or tunnels.

9. SSH sessions and the terminal

> **What is SSH?** SSH is an encrypted connection to a server that gives you access to its command line (terminal). Everything you type and everything the server sends back is encrypted, passwords included.

To connect, double-click a host. If the identity contains no password, the program shows a **Log in to …** window with **Username** and **Password** fields. Selecting **Save to identity** saves the password you enter in

the host's identity.

Host key on first connection

> **What is a host key and fingerprint?** Every SSH server has its own key that identifies it, much like a seal. The fingerprint is a short form of that key, e.g. `SHA256:7lrypwr…`. On the first connection the program shows the fingerprint and remembers it. On later connections it checks that it is talking to the same server and not an impostor.

The first time you connect to a server, an **Unknown host key** window appears with the key type (**Key type**) and the fingerprint (**Fingerprint**). Compare the fingerprint with the one provided by the server's administrator and click **Trust and connect**. Trusted keys are listed in **Keychain → Known hosts**.

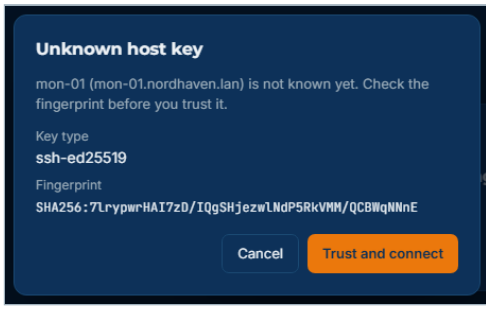


Figure 10. Unknown host key on the first connection to mon-01 (sample data)

Changed host key

When a server presents a different key from the one remembered, the program holds the connection and shows **The host key has changed** with the new (**New fingerprint**) and previous (**Previous fingerprint**) fingerprints. Usually this means the server was reinstalled, but it can also mean someone is trying to intercept the connection.

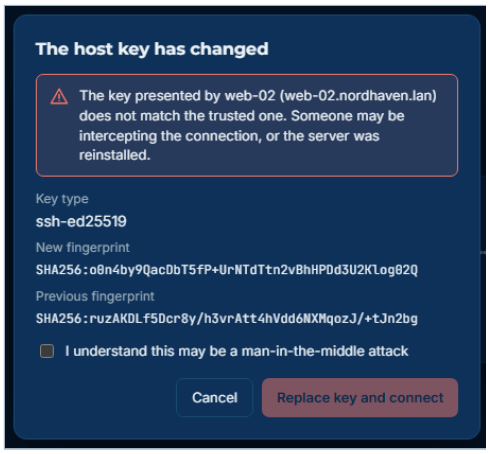


Figure 11. Changed host key warning for web-02 (sample data)

> **Note.** Do not accept a changed key without checking. Ask the server's administrator whether the key was changed and compare the new fingerprint. Only then select **I understand this may be a man-in-the-middle attack** and click **Replace key and connect**. If in doubt, click **Cancel**.

Working in the terminal

The terminal works like an ordinary server console. `Ctrl`+`Shift`+`C` copies text selected with the mouse, and `Ctrl`+`Shift`+`V` pastes. `Ctrl`+`Shift`+`F` opens find in the terminal contents. The right-click menu contains **Copy**, **Paste**, **Select all**, **Find**, **Clear scrollback**, plus **Snippets** and **Type password**.

When you paste several lines, the program asks for confirmation by default, because each line may run as a separate command. The **Multi-line paste** setting changes this behavior (see "Settings").

Server statistics bar

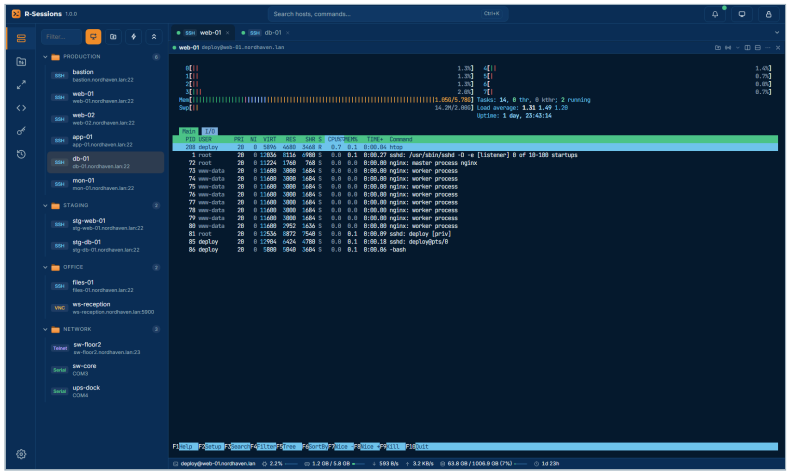


Figure 12. The web-01 terminal running htop, with the server statistics bar at the bottom of the window (sample data)

The bar at the bottom of the window shows the address of the current session and, for Linux servers, data refreshed every few seconds: CPU usage, used and total memory, download and upload speed, usage of the root disk (`/`) and server uptime. The data is read over a separate channel of the same connection, so it never appears in the terminal or the command history. The **Show server stats** field in the host editor turns the bar on and off. On servers other than Linux the bar shows only the address.

Split panes and broadcast

One tab can hold several panes, e.g. a log viewer next to htop. Buttons on the pane bar split it to the right (**Split right**, `Alt` + `Shift` + `=`) or down (**Split down**, `Alt` + `Shift` + `-`); the new pane connects to the same host. **Split with another host…** in the **More actions** menu (three dots) opens a different host in the new pane. `Ctrl` + `Shift` + `W` closes a pane.

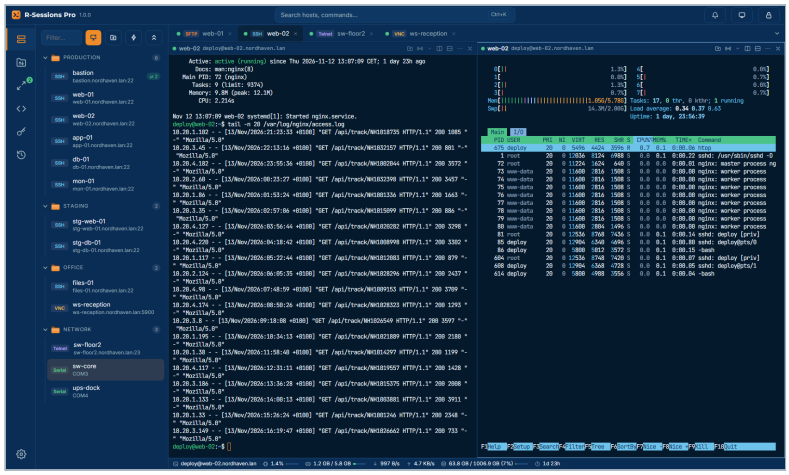


Figure 13. The web-02 tab split into two panes: the web server log and htop (sample data)

Broadcast sends what you type to several sessions at once, e.g. to run the same command on three servers. The **Broadcast input (to selected panes)** button on the pane bar turns it on, and the arrow next to it (**Choose broadcast targets**) selects the sessions. While broadcast is on, a "Broadcast on: …" bar with the list of sessions appears above the terminal, and the receiving tabs carry a separate marker.

Lost connection

When a connection drops (e.g. a server restart or a network outage), the dot on the tab turns red, and a **Connection lost:** bar appears at the bottom of the pane with the reason and the **Reconnect** (`Enter`) and **Close** (`Esc`) buttons. The event is also added to the bell. The terminal contents stay, so after reconnecting you can see what happened before.



Figure 14. The Connection lost bar after the connection to web-02 dropped: Reconnect (Enter) and Close (Esc) (sample data)

With **Settings** → **Sessions** → **Reconnect automatically** enabled, the program tries to reconnect on its own up to three times: after 2, 5 and 10 seconds. While the vault is locked, it does not reconnect automatically.

Local shell

The **Open local shell…** command in search opens this computer's command line (e.g. PowerShell or bash) on a tab. A shell you use often can be saved as a host with the **Local shell** protocol, with **Shell** and **Start folder** fields. Such hosts do not count toward the Free edition limit.

10. Files (SFTP)

> **What is SFTP?** SFTP transfers files over the same encrypted SSH connection the terminal uses. It needs no separate FTP server or extra passwords: if you can sign in to a server over SSH, you can also browse and transfer its files.

Files open on a tab with two panes. To open it, use:

- a click on a host in the **SFTP** section (the second icon in the left bar);
- the **Open SFTP** command in the host menu or on the terminal's pane bar;
- `Ctrl` + `Shift` + `O` in the terminal: the right pane then shows that session's server, and the **Follow terminal** button goes to the folder you are in in the terminal.

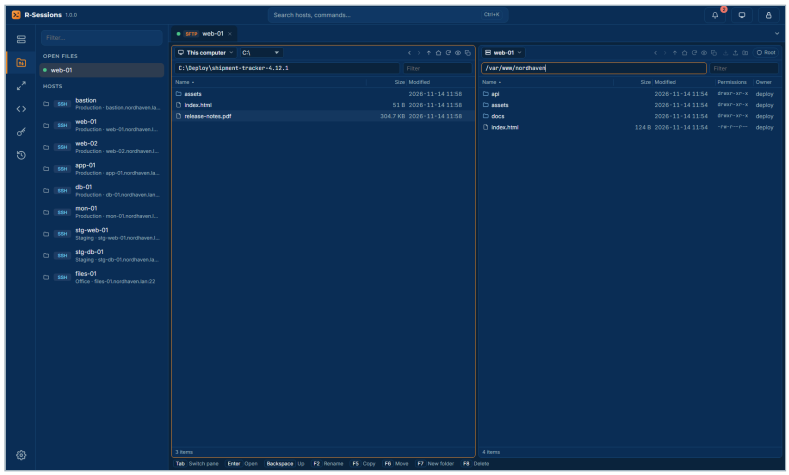


Figure 15. The files tab: a folder on this computer on the left, the /var/www/nordhaven folder on the web-01 server on the right (sample data)

Each pane can show this computer (**This computer**, with drive selection on Windows) or any SSH host, so you can also copy files between two servers. Above the list are the **Back**, **Forward**, **Up**, **Home**, **Refresh**, **Show hidden files** and **Copy path** buttons, and in a server pane also **Download to…**, **Upload files…**, **Upload folder…** and **Root**. You can type a path in the address field, and the **Filter** field narrows the list. The **Permissions** and **Owner** columns show file permissions and owners on the server.

Key	Action	Key	Action
`Tab`	switch to the other pane	`F5` / `F6`	copy / move selected items to the folder in the other pane
`Enter`	open a folder or file	`F7`	new folder
`Backspace`	parent folder	`F8`	delete (with confirmation)
`F2`	rename		

The right-click menu also includes **Edit**, **Edit with…**, **Download to…**, **Copy path**, **Rename**, **Permissions…** and **Delete**, and on an empty area of the list **New folder**, **New file**, **Upload files…**, **Upload folder…** and **Refresh**.

Copying files

1. In one pane, open the destination folder; in the other, select files or folders (click, with `Ctrl` or `Shift`).
2. Press `F5` (copy) or `F6` (move). You can also drag items to the other pane.
3. In the **Copy** window, check where the files come from (**From**) and where they go (**To**), choose an action in **If a file exists** and click **Copy**.

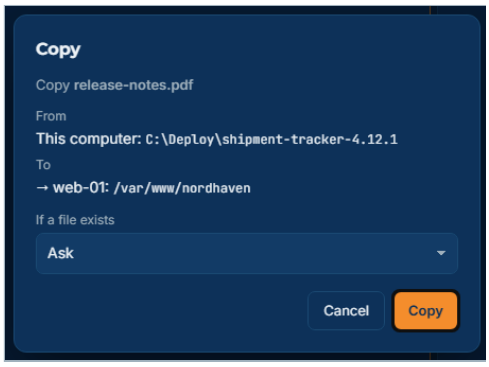


Figure 16. The Copy window: copying release-notes.pdf to the web-01 server (sample data)

If a file exists	What happens when a file with that name already exists in the destination folder
Ask	the program asks for each such file (with an **Apply to all** option), showing the size and date of both versions
Overwrite	replaces the existing file
Skip	keeps the existing file and does not copy it
Keep both	copies the file under a changed name
Resume	completes a previously interrupted transfer of this file

> **Tip.** Files cannot be dragged into a pane from Windows Explorer or a Linux file manager. Use the **Upload files…** and **Upload folder…** buttons, and **Download to…** for the other direction.

Transfer queue

Transfers run in the background, and their progress is shown in the **Transfers** panel below the file lists. By default it shows the transfers of this tab; **All transfers** shows all of them, and **Clear finished** removes completed ones from the list. Each transfer has **Pause**, **Resume** and **Cancel** buttons (the menu also has **Cancel and keep partial**, which keeps the partially transferred file for resuming later). When a transfer finishes, a short message appears, e.g. **Upload finished**, which also stays in the bell.

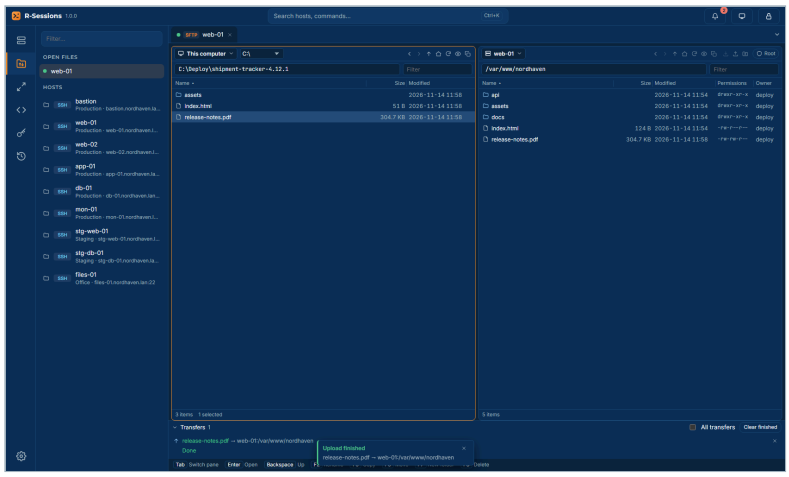


Figure 17. The Transfers panel and the Upload finished message after a file was sent (sample data)

When the connection drops, a transfer shows "Waiting for connection" and resumes after reconnecting. Large files (64 MiB and over) and resumed transfers are verified with a SHA-256 checksum, and many small files are packed into a single tar archive if both sides have tar (settings in **Settings** → **Files**).

Editing files on the server

`Enter`, a double-click or **Edit** opens a text file in the built-in editor, in a separate window. Saving the file in the editor sends it back to the server. In **Settings** → **Files** → **Editor** you can choose a different program instead, and **Edit with…** picks a program for a single file. Files being edited in an external program are listed on the **Editing** bar of the files tab; the **Done editing** button ends editing.

Root mode (sudo)

> **What is sudo and Root mode?** On Linux servers, a regular user cannot change system files, such as configuration in `/etc`. The sudo command lets you perform an operation with administrator (root) privileges after entering a password. Root mode does the same in the file pane.

1. In the server pane, click the **Root** button above the file list.
2. If no sudo password is saved, the program shows the **Root mode on …** window. Enter the **Sudo password**; **Save to identity** saves it in the host's identity.
3. The **Root** button is highlighted and the pane works with root privileges. Click it again to turn the mode off.

> **Note.** In Root mode you can delete or overwrite any system file. Turn it on only for the changes you need to make.

11. Tunnels

> **What is a tunnel?** A tunnel forwards a connection from a port on your computer to a service that is reachable only from inside the company network, e.g. a database on the db-01 server. The traffic goes through an encrypted SSH connection to a server you have access to. Example: the tunnel `127.0.0.1:15432 → db-01.nordhaven.lan:5432` lets a database client on your computer connect to address `127.0.0.1`, port `15432`, while it actually talks to the database on db-01. A port is the number of a service on a computer, e.g. 5432 for PostgreSQL, 3000 for Grafana.

Type	Where it listens	Typical use
Local	a port on your computer; connections reach a service as seen from the SSH server	a database, a web panel or another service in the server's network
Remote	a port on the SSH server; connections reach a service as seen from your computer	making a service from your network available to the server, e.g. a local test server
Dynamic	a port on your computer acting as a SOCKS5 proxy server	a browser set to use this proxy opens pages as if it were in the server's network

Creating a tunnel

1. In the **Tunnels** section, click **New tunnel** (or choose **New tunnel…** from an SSH host's menu).
2. Choose the **Type**: **Local**, **Remote** or **Dynamic**.
3. In **Host**, choose the SSH server the traffic should go through, and enter a **Name**.
4. Set **Listen on** and **Listen port**, and for Local and Remote tunnels also **Target host** and **Target port**.
5. The diagram and the sentence below it describe the tunnel in words, e.g. "Connections to 127.0.0.1:16379 on this computer reach app-01.nordhaven.lan:6379 from bastion." Check them.
6. In **Start**, choose when the tunnel should run, and click **Save**.



Figure 18. New Local tunnel redis-cache: port 16379 on this computer leads through bastion to app-01, port 6379 (sample data)

Field	Meaning
Listen on	**This computer only (127.0.0.1)**: only this computer can use the tunnel (recommended); **All interfaces (0.0.0.0)** or **Custom**: the tunnel is also available to other computers on the network, which the program warns about
Listen port	the port number on which the tunnel accepts connections; in a Remote tunnel, **Let the server choose** lets the server pick a free port
Target host, **Target port**	the destination service, as the other end of the tunnel sees it (for Local: the SSH server)
Start	**Manually**: with the switch; **With the host**: together with the first session of this host; **Always on**: right after the vault is unlocked

Note. A **Dynamic** tunnel listening on all interfaces lets anyone who can reach that port use the server as a proxy. Unless you clearly need it, keep **This computer only (127.0.0.1)**.

Tunnel list and starting tunnels

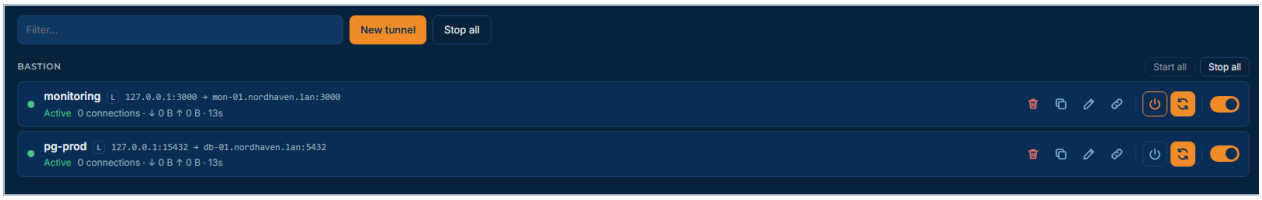


Figure 19. The Tunnels section: the monitoring and pg-prod tunnels of the bastion host, both active (sample data)

Tunnels are grouped by host. Each one shows its type (**L**, **R**, **D**), addresses, state (**Active**, **Stopped**, **Starting…**, **Reconnecting…**, **Needs attention**, **Failed**), number of connections, data transferred and running time. The switch on the right starts and stops the tunnel. The other buttons are, in order: **Delete**, **Duplicate**, **Edit**, **Copy address** (an address to paste into another program), **Autostart** (changes the **Start** field) and **Reconnect** (reconnects after a drop). **Start all** and **Stop all** work for one host, and **Stop all** at the top for all tunnels.

The **Needs attention** state means the tunnel is waiting for a decision, e.g. accepting a host key or entering a password. The number of running tunnels is shown on the **Tunnels** icon in the left bar, and a ⇄ marker with the number of its tunnels appears next to a host in the list. Tunnels can also be started from `Ctrl` + `K` search. The Free edition can save 2 tunnels.

12. Snippets

> **What is a snippet?** A snippet is a saved command or short script that the program types into the terminal for you, e.g. checking a service's status or deleting old backups. A snippet can have variables whose values the program asks for before running it.

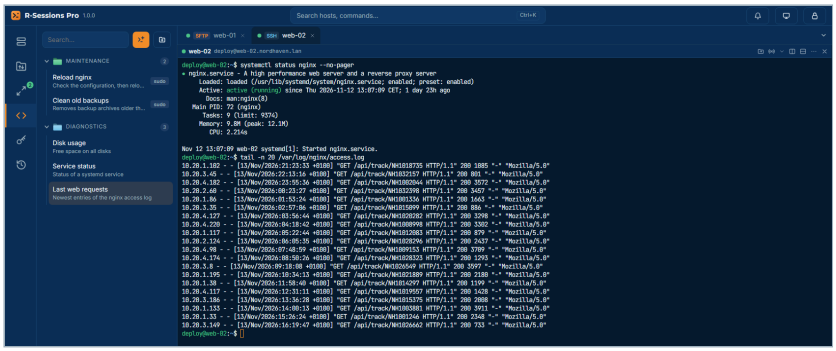


Figure 20. The Snippets section with the Maintenance and Diagnostics packages, and the result of the Service status snippet in the web-02 terminal (sample data)

Snippets are in the **Snippets** section (the <> icon), grouped into packages. The **Search…** field filters them. **New snippet** adds a snippet, and **New package** adds a package (name and color). A snippet marked **sudo** runs with administrator privileges.

Creating a snippet

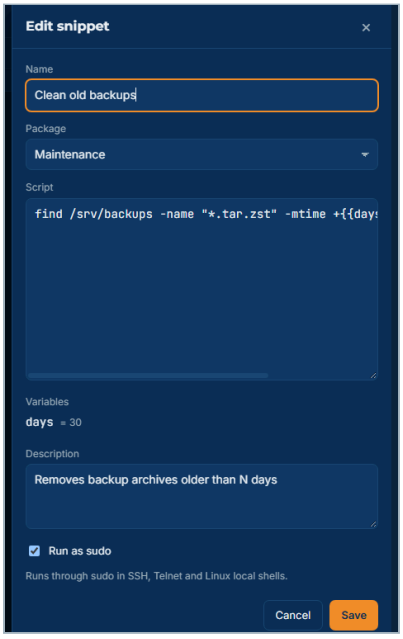


Figure 21. Editing the Clean old backups snippet with the days variable and Run as sudo (sample data)

1. Click **New snippet**.
2. Enter a **Name**, choose a **Package** and type the commands in **Script**, one per line.
3. Write the parts that should change as variables: `{{name}}` or `{{name:default value}}`, e.g. `-mtime +{{days:30}}`. The **Variables** field shows the variables found.
4. Optionally add a **Description** and select **Run as sudo**. Click **Save**.

Running snippets

A snippet runs in the active terminal session, and with broadcast on, also in the selected sessions. To run one, use:

- a double-click or `Enter` in the snippet list (`F2` edits, `Delete` deletes);
- the **Snippets** submenu in the tab or terminal menu;
- `Ctrl`+`Shift`+`P` (snippet list) or `Ctrl`+`K` search;
- the host's **Startup snippet** field: automatically after connecting.

If the snippet has variables, a **Run snippet** window appears with the session name and the variable fields. Enter the values and click **Run**.

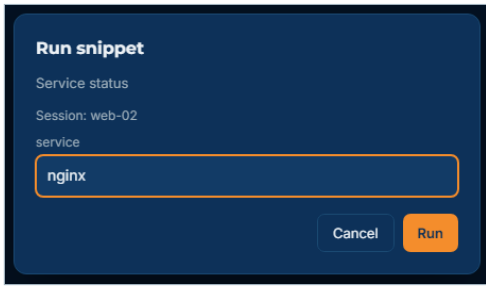


Figure 22. The Run snippet window: the value of the service variable before running in the web-02 session (sample data)

Run as sudo runs the script through sudo. When sudo asks for a password, the program types it in itself if one is saved: first the host's user password with **Use for sudo**, then the identity's **Sudo password**, and finally the identity's sign-in password. Without a saved password, you type it in the terminal. Sudo works in SSH and Telnet sessions and in Linux local shells; it does not work over a serial port or in Windows shells.

13. Telnet and serial port

> **What is Telnet and why is it unencrypted?** Telnet is an older way than SSH to connect to a device's command line. It sends everything, including the login and password, as plain text, so anyone listening on the network can read it. It is still used by older switches, routers and console servers that have no SSH.

A Telnet host is created like an SSH host, with **Protocol** set to **Telnet** (default port 23). **Raw mode** turns off Telnet option negotiation; it is needed for console servers and raw TCP connections. A Telnet session shows **Unencrypted** on the pane bar and the statistics bar.

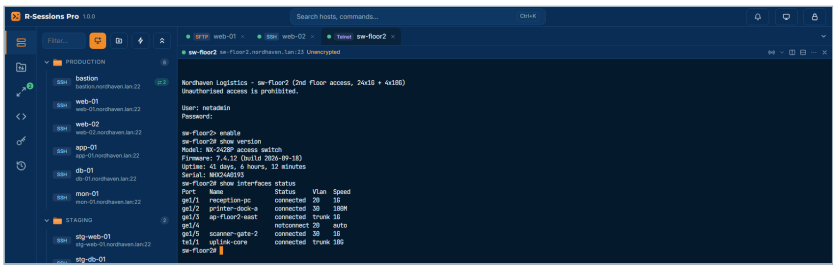


Figure 23. A Telnet session with the sw-floor2 switch, marked Unencrypted (sample data)

> **Note.** Use Telnet only on an isolated, trusted network. If the device supports SSH, choose SSH.

Serial port

> **What is a serial port?** A serial port (COM on Windows, e.g. `/dev/ttyUSB0` on Linux) is a connector for plugging a console cable directly into a device, such as a switch or a UPS. It is useful when the device has no network address yet or the network is down. Modern computers use a USB adapter for this.

1. Connect the console cable and add a host with **Protocol** set to **Serial**.
2. In **Serial port**, choose a port from the list or type its name. The button next to it refreshes the port list.
3. Set the parameters according to the device documentation and click **Save**.

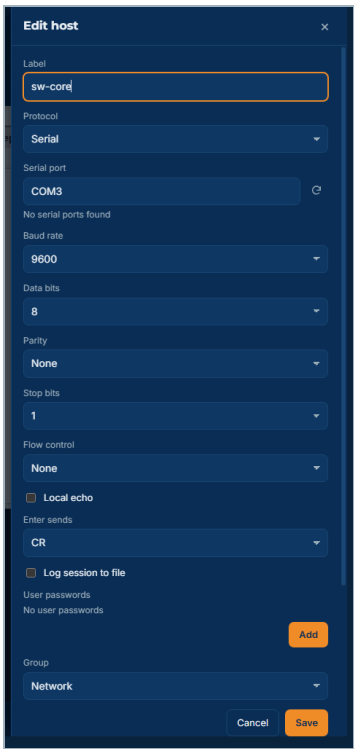


Figure 24. Editing the sw-core host connected to port COM3 (sample data)

Field	Meaning
Baud rate	transmission speed, e.g. 9600 or 115200
Data bits, **Parity**, **Stop bits**	character format; most often 8, None, 1
Flow control	**None**, **Hardware (RTS/CTS)** or **Software (XON/XOFF)**
Local echo	shows the characters you type when the device does not echo them
Enter sends	line ending: **CR**, **LF** or **CR+LF**
Log session to file	records the session to a file

Defaults for new Serial and Telnet hosts are set in **Settings → Serial & Telnet**. The **Send Break** command in the tab menu sends a Break signal, needed e.g. for password recovery on some devices.

14. VNC

> **What is VNC?** VNC shows another computer's desktop in a window and lets you control it with the mouse and keyboard, e.g. to help an employee at the reception desk. A VNC server must run on the remote computer. A VNC connection on its own is usually not encrypted, which is why R-Sessions can route it through an SSH server.

A VNC host is created with **Protocol** set to **VNC** (default port 5900). Only the password from the identity is used (**Only the password is used**). The other fields:

Field	Meaning
Connect through	**Direct connection** or an SSH host through which the connection is encrypted; the address and port are then entered as that host sees them
Scaling	**Fit to window**, **Resize remote screen** (changes the remote screen resolution), **Actual size**
Quality	**Auto**, **Low**, **Medium** or **High**; lower quality for a slow link
View only	view without control
Clipboard	shared clipboard with the remote computer

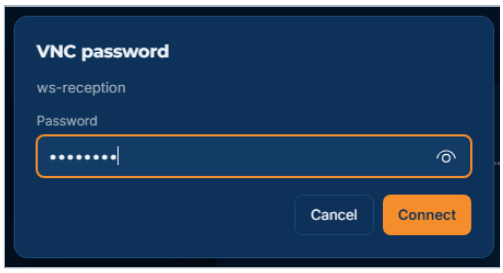


Figure 25. The VNC password window for the ws-reception computer (sample data)

If the identity contains no password, the program asks for it in the **VNC password** window.

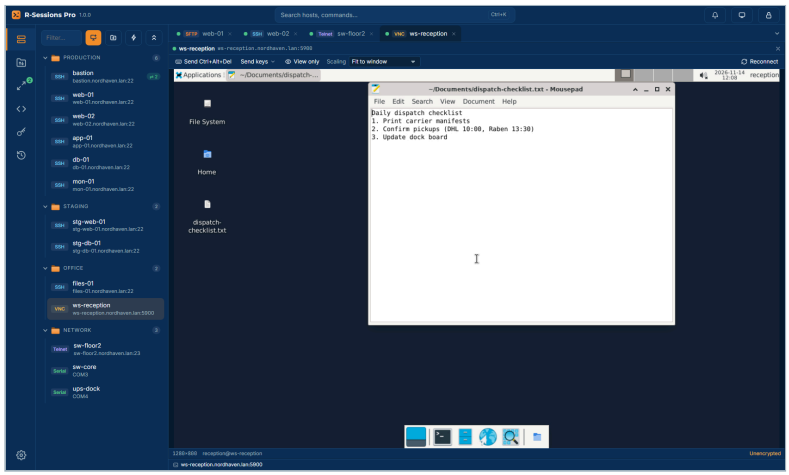


Figure 26. A VNC session with the ws-reception computer and the toolbar above the desktop (sample data)

The toolbar above the desktop contains **Send Ctrl+Alt+Del**, **Send keys** (Ctrl+Esc, Alt+Tab, Alt+F4, the Windows key, Print Screen), **View only**, **Scaling** and **Reconnect**. Below the desktop you can see the resolution, the remote desktop name and the encryption state: **Unencrypted** or **Encrypted (SSH)**.

> **Note.** A **Direct connection** sends the screen image and keystrokes without encryption. Outside a trusted local network, choose an SSH host in **Connect through**.

15. Connection history

The **History** section (the clock icon in the left bar) shows connections grouped by day: type, name, user and address, start time, and duration or "in progress" for active ones, with **Failed** for unsuccessful ones. The **Search…** field searches the entries, and the **All protocols** list narrows them to one connection type.



Figure 27. Connection history with the Connect, Edit host and Remove entry buttons (sample data)

Connect (or a double-click) connects again, **Edit host** opens the host editor, and **Remove entry** deletes the entry. A Quick connect entry reopens Quick connect with the same address. The history retention period (Off, 30, 90 or 365 days) and the **Clear history** button are in **Settings** → **Sessions**. History is stored in the vault, encrypted.

16. Notifications

The bell in the title bar collects the program's messages: dropped connections, finished transfers, tunnel problems, license loading, available updates. Short on-screen messages disappear by themselves but stay in the bell. The number on the bell shows new entries, and red means current errors.

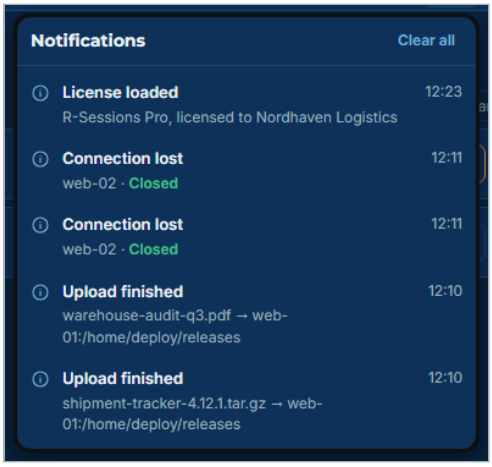


Figure 28. The bell: license loaded, connection lost and file upload finished (sample data)

- Clicking an entry takes you to its source, e.g. the session tab, the tunnel list or the license settings.
- Repeated events are merged into one entry with a counter (e.g. ×3), and issues already resolved are marked **Resolved** or **Closed**.
- **Clear all** removes all entries.
- Notifications are kept in memory only, until the program closes, and are not visible while the vault is locked.

Part III. Administration

17. Installation and updates

Requirements

- Windows 10 or 11, 64-bit (the Microsoft Edge WebView2 component is included in these systems).
- 64-bit Linux (x64): deb and rpm packages, or an AppImage file for other distributions.
- Installation is done once per computer; each system user has their own vault.

Download and installation

Installers can be downloaded from `https://r-sessions.com/download`. `X.Y.Z` in the file name is the version number.

System	File	Installation
Windows 10 / 11	`R-Sessions_X.Y.Z_x64-setup.exe`	run the file and follow the installer
Windows, company deployment (GPO, Intune, SCCM)	`R-Sessions_X.Y.Z_x64_en-US.msi`	MSI package, e.g. `msiexec /i R-Sessions_X.Y.Z_x64_en-US.msi /qn`
Ubuntu, Debian, Linux Mint	`R-Sessions_X.Y.Z_amd64.deb`	`sudo apt install ./R-Sessions_X.Y.Z_amd64.deb`
Fedora, RHEL, Rocky Linux, AlmaLinux	`R-Sessions-X.Y.Z-1.x86_64.rpm`	`sudo dnf install ./R-Sessions-X.Y.Z-1.x86_64.rpm`
Other Linux distributions	`R-Sessions_X.Y.Z_amd64.AppImage`	`chmod +x` on the file, then run it

The `.exe` and `.msi` files are digitally signed by **RALKOM RAFAŁ LESIŃSKI**. To check the signature in Explorer, right-click the file, choose **Properties** and open the **Digital Signatures** tab; in PowerShell, use `Get-AuthenticodeSignature` (the result should be `Valid`). Do not run an installer without a valid signature.

Updates

Once a day the program checks for a new version (the **Check for updates automatically** setting in **Settings** → **About**, on by default). A new version is announced by an entry in the bell. In **Settings** → **About**, **Check now** checks immediately, and **Install and restart** downloads the update and asks before installing, listing what will be closed (sessions, transfers). Downloading does not interrupt your work. After the restart, the bell shows an "Updated to R-Sessions …" entry.

Every update is checked against the R-Sessions update signing key before installation. An update without a valid signature is not installed (message **The update is not signed correctly and was not installed.**). If installation fails, the **Open download page** button opens the release page for a manual download.

Where the program keeps its data

Path	Contents
Windows: `%APPDATA%\rsessions\R-Sessions\data\` Linux: `~/.local/share/r-sessions/`	the vault `rsessions.db` (encrypted) and the default session log folder `logs`
Windows: `%LOCALAPPDATA%\rsessions\R-Sessions\data\edit\` Linux: `~/.local/share/r-sessions/edit/`	working copies of files opened for editing in an external program

Uninstalling the program does not delete the vault. To move your data to another computer, use configuration export (see "Import and export"), not a copy of the vault file.

18. Keychain: identities, keys, known hosts

The **Keychain** section (the key icon in the left bar) has three tabs: **Identities**, **Keys** and **Known hosts**.

Identities

An identity combines a username with a sign-in method. Many hosts can use one identity, so changing a password in one place works for all of them.



Figure 29. The Identities tab: an identity with a key and an identity with a password (sample data)

1. Click **New identity**.
2. Enter an **Identity label** (the name in the list) and a **Username**.
3. In **Authentication**, choose **Password**, **Key** or **SSH agent**. With a password, the **Password** field can stay empty: the program asks for the password when connecting.
4. Optionally enter a **Sudo password**, used in file Root mode and in snippets run through sudo. Click **Save**.

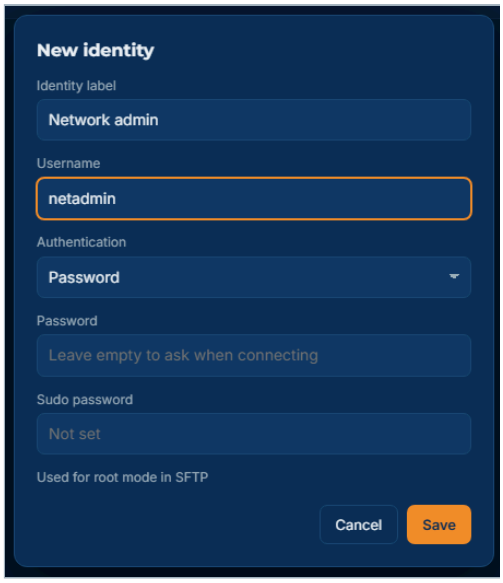


Figure 30. The New identity window (sample data)

View password shows the saved password after you enter the vault password again. **Edit** and **Delete** change and delete the identity.

SSH keys

> **What is an SSH key?** An SSH key is a pair of files that replaces a password. The private key stays with you (in the vault), and the public key goes on the server, in the user's `~/.ssh/authorized_keys` file. When you sign in, the program proves to the server that it holds the private key without sending it. A key cannot be sniffed on the network or guessed like a password.



Figure 31. The Keys tab: the deploy@nordhaven key with its fingerprint (sample data)

- **Generate key** creates a new key: enter a **Key label** and choose the **Algorithm**: **Ed25519** (recommended), **RSA 3072**, **RSA 4096** or **ECDSA P-256**. An RSA key takes a few seconds to generate.
- **Import key** adds an existing private key in OpenSSH format: paste it into **Private key (paste)** or select the file with **Choose file…**. An encrypted key requires a **Key passphrase**.
- **Copy public key** copies the public key to the clipboard. Paste it as a new line in the `~/.ssh/authorized_keys` file on the server.
- **Export private key…** saves the private key to a file after you enter the vault password, optionally encrypted with a new passphrase (**Export passphrase (optional)**).

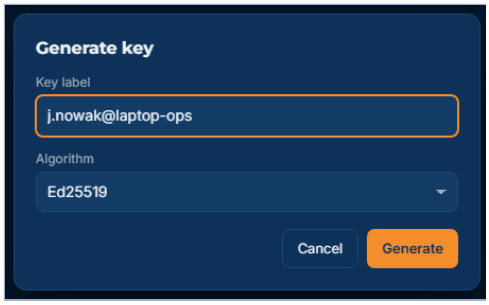


Figure 32. The Generate key window (sample data)

SSH agent and agent forwarding

> **What is an SSH agent?** An SSH agent is a background program that holds unlocked keys and signs sign-ins with them at the request of other programs. On Windows this is the OpenSSH Authentication Agent service or Pageant; on Linux, the agent pointed to by the `SSH_AUTH_SOCK` variable.

An identity with **Authentication** set to **SSH agent** signs in with keys from the system agent. On Windows, R-Sessions looks for the OpenSSH service first, then for Pageant.

The **Agent forwarding** field in the host editor makes keys from the Keychain and from the system agent available to the server for the duration of the session. This lets you sign in onward from that server (e.g. `git pull` or `ssh` to another server) without copying your private key to the server.

> **Note.** The administrator of a server with agent forwarding enabled can use your keys during your session. Enable this option only on servers you trust. To connect through an intermediate server, a jump host is the better choice (next chapter).

Known hosts

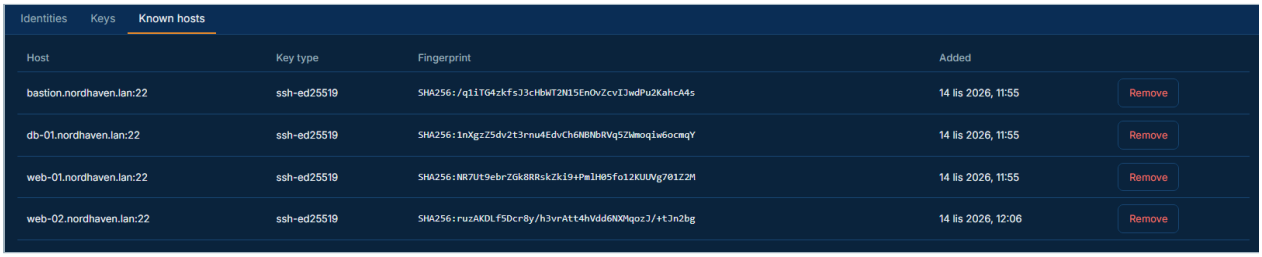


Figure 33. The Known hosts tab: trusted server keys (sample data)

The list contains trusted server keys: host and port, key type, fingerprint and the date added. **Remove** deletes a trusted key, e.g. after a planned server reinstall; on the next connection the program asks about the key as it does the first time.

19. Jump hosts

> **What is a jump host (bastion)?** In many companies, internal servers are not reachable directly. You first connect to one specially hardened server, called a bastion or jump host, and only through it to the others. R-Sessions does this automatically: just point to the bastion in the host settings. The connection to the target server is encrypted all the way from your computer to the target, and the bastion only passes encrypted data along.

1. Add the bastion as a regular SSH host (e.g. **bastion**, `bastion.nordhaven.lan`) with its own identity.
2. Open the target server's editor and choose the bastion in **Jump host**. The connection route appears below the field, e.g. "Connects via bastion → app-01".
3. Save the host and connect to it as usual.

A bastion can have its own jump host, so the route can lead through several servers. Each server on the route signs in with its own identity, and its key is verified separately: the **Unknown host key** window then shows a **Jump host:** note with the bastion's name.

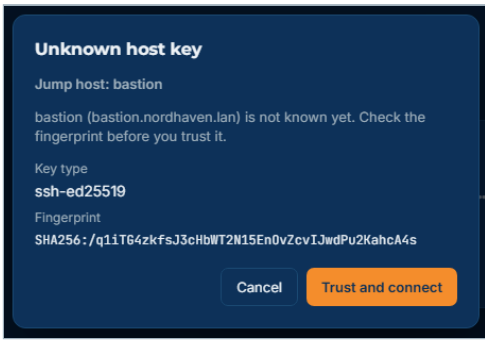


Figure 34. Verifying the bastion's key when connecting to a server behind it (sample data)

SFTP files, tunnels and snippets also work through a jump host. VNC hosts specify the SSH server in **Connect through**.

20. Import and export

All commands are in **Settings** → **Import & export**.

Import from OpenSSH

OpenSSH config → **Import…** reads the `~/.ssh/config` file (with files pulled in by `Include`) and shows a list of hosts to import: name, protocol, address, user, key and remarks. Hosts that already exist in the program are marked **Already exists**. Select the hosts, choose a **Group** (by default a new "Imported from OpenSSH" group, an existing group or **No group**) and confirm. Keys referenced by `IdentityFile` go into the Keychain; an encrypted key needs its passphrase in the **Key passphrases** part (an empty field imports the hosts without that key). `ProxyJump` becomes a jump host. Port forwards and `Match` blocks are not imported, and the list says so.

Import from PuTTY

PuTTY sessions → **Import…** reads PuTTY sessions (on Windows from the registry, on Linux from the PuTTY settings directory) with the SSH, Telnet and Serial protocols. `.ppk` keys must first be converted to OpenSSH format with PuTTYgen and imported in the Keychain. PuTTY proxy settings are skipped.

Exporting and importing the R-Sessions configuration

Export moves your configuration to another computer or creates a backup of it. The `r-sessions-YYYY-MM-DD.rsexport` file is encrypted with its own password.

1. In **R-Sessions configuration**, click **Export…**.
2. Select what to export: **Hosts**, **Identities and keys**, **User passwords**, **Snippets**, **Tunnels**, **Known hosts**, **Settings**. User passwords and tunnels require hosts.
3. Enter an **Export password** (at least 10 characters, different from the vault password), repeat it in **Confirm password**, enter the **Vault password** and click **Export**.

On the other computer, click **Import…**, select the file and enter the **File password**. A table shows, for each category, the number of new items, existing items and conflicts; in **Existing items**, choose **Skip** (keep the current ones) or **Replace** (replace with those from the file).

> **Note.** The export file contains passwords and private keys. They are protected by the export password, so choose a strong one and pass it on through a different channel than the file. The license is not included in the export: load the license file again on the new computer. In the Free edition, import adds hosts and tunnels only up to the limit and reports how many it skipped.

21. Settings

Settings open with the gear at the bottom of the left bar. The pages are listed on the left side of the window.

Security

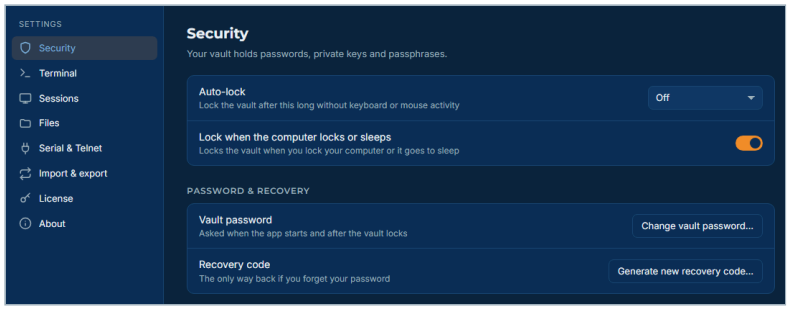


Figure 35. Settings → Security (sample data)

Setting	Meaning
Auto-lock	locks the vault after inactivity: Off, 5, 15, 30 or 60 minutes
Lock when the computer locks or sleeps	locks when the screen is locked or the computer sleeps (on Linux, requires systemd-logind)
Change vault password…	changes the vault password; requires the current password
Generate new recovery code…	a new recovery code after you enter the vault password; the previous one stops working

Terminal

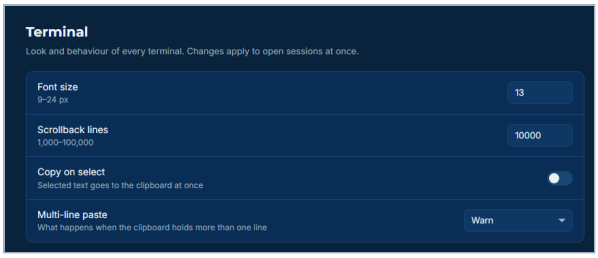


Figure 36. Settings → Terminal (sample data)

Setting	Meaning
Font size	terminal font size, 9–24 px
Scrollback lines	how many lines you can scroll back, 1,000–100,000
Copy on select	selecting text copies it to the clipboard right away
Multi-line paste	pasting several lines: **Paste directly** (no prompt), **Warn** (prompt, default), **Edit before pasting** (preview to adjust)

Sessions

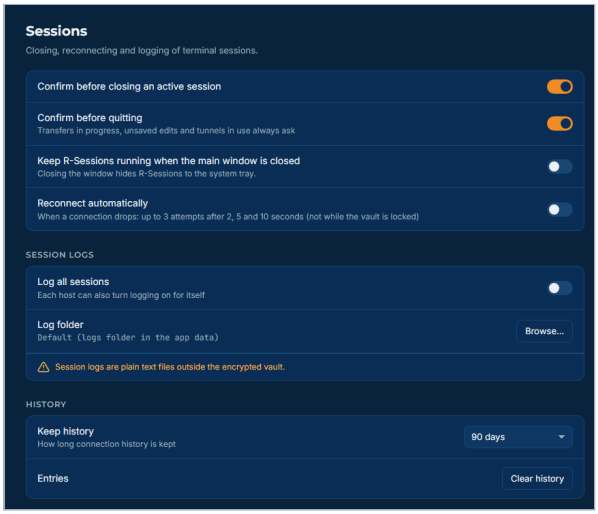


Figure 37. Settings → Sessions (sample data)

Setting	Meaning
Confirm before closing an active session	asks before closing a tab with an active session
Confirm before quitting	asks before closing the program; with running transfers, unsaved edits and tunnels in use it always asks
Keep R-Sessions running when the main window is closed	closing the window hides the program in the notification area next to the clock (Windows); on Linux the window is minimized
Reconnect automatically	up to 3 reconnect attempts after 2, 5 and 10 seconds
Log all sessions, **Log folder**	logs of all sessions and the log folder (see "Session logs")
Keep history, **Clear history**	connection history retention period and clearing it

Files

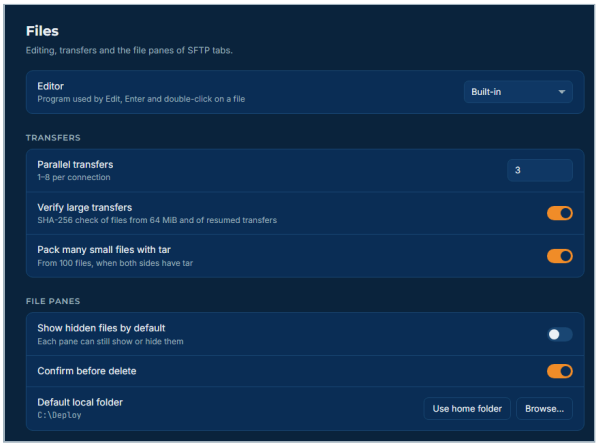


Figure 38. Settings → Files (sample data)

Setting	Meaning
Editor	program for editing files: **Built-in**, one of the detected programs or **Other…**
Parallel transfers	number of files transferred at the same time per connection, 1–8
Verify large transfers	SHA-256 checksum verification of files of 64 MiB and over and of resumed transfers
Pack many small files with tar	from 100 files, packs them into one archive when both sides have tar
Show hidden files by default	shows hidden files in new panes
Confirm before delete	asks before deleting files
Default local folder	the folder shown in the **This computer** pane; **Use home folder** restores the home folder

Serial & Telnet

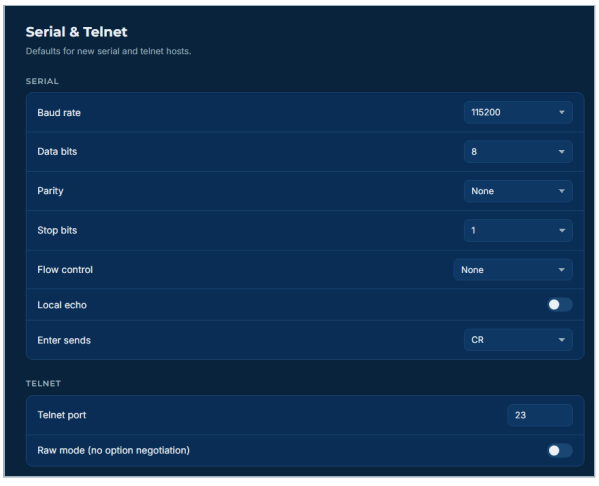


Figure 39. Settings → Serial & Telnet (sample data)

Defaults for new hosts: serial port parameters (**Baud rate**, **Data bits**, **Parity**, **Stop bits**, **Flow control**, **Local echo**, **Enter sends**), plus **Telnet port** and **Raw mode (no option negotiation)**. Changes do not affect hosts already saved.

Import & export



Figure 40. Settings → Import & export (sample data)

Import from OpenSSH and PuTTY, and export and import of the R-Sessions configuration (see "Import and export").

License and About

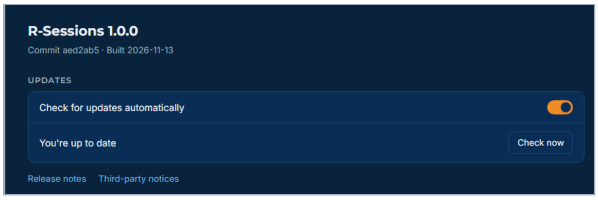


Figure 41. Settings → About: program version and updates (sample data)

License is described in "License". **About** shows the program version, the build number (**Commit**) and build date, the update settings, and the **Release notes** (list of changes) and **Third-party notices** (licenses of the open source libraries used) links.

22. License

Without a license, the program runs as R-Sessions Free. **Settings → License** then shows how many hosts and tunnels are saved, e.g. "13 of 15 hosts, 2 of 2 tunnels", and the **Load license file** and **Get Pro** (purchase page) buttons.

When the limit is reached, adding a host or tunnel (also through **Duplicate**, import or **Save as host**) opens the **Free version limit reached** window with the **Close**, **Load license file** and **Get Pro** buttons. Existing hosts and tunnels keep working and can be edited without restriction, and you can connect to new servers with Quick connect.

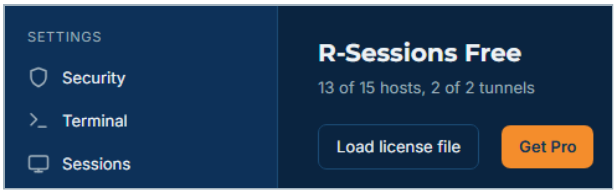


Figure 42. Settings → License in the Free edition (sample data)

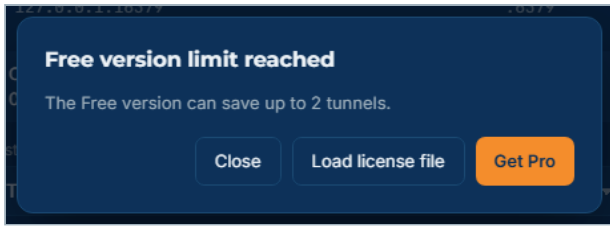


Figure 43. The Free edition limit window when trying to save a third tunnel (sample data)

Loading a license

1. Save the license file you received, e.g. `R-Sessions-RS-2026-000123.rslicense`, on your computer.
2. In **Settings → License**, click **Load license file** and select the file.
3. The program checks the license signature on your computer, without connecting to the internet, and stores it in the vault. A **License loaded** entry appears in the bell.

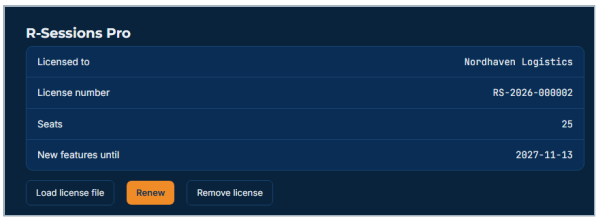


Figure 44. Settings → License with a Pro license (sample data)

Field	Meaning
Licensed to	the license owner: a person or a company
License number	the license number, needed for renewal and when contacting us
Seats	number of seats; one seat is one person on any number of their computers
New features until	until this date the license covers new Pro features; it is **not** an expiry date, the program keeps working afterwards, and "(ended)" appears next to the date

- **Renew** opens the license renewal page; only the license number is added to the address. Load the renewed file with **Load license file**. When you load an older file of the same license, the program asks for confirmation.
- **Remove license** removes the license from this computer (after confirmation); the program switches to the Free edition, and no data is deleted.
- The license is stored in the vault, encrypted. Load the same file on every computer you work on.

Blocked licenses

A license can be blocked, e.g. after a refund or a public leak of the file. The program knows the list of blocked licenses built into its version, and once a day, after the vault is unlocked, it downloads a signed update of that list with a plain request, without sending any data. A blocked license works like the Free edition, and **Settings → License** shows the message "This license has been disabled. Please contact kontakt@ralkom.pl for a replacement." In that case, contact RALKOM (see "Technical support").

23. Session logs

A session log is a text file with what appeared in the terminal. It is useful for documenting work or analyzing a failure afterwards.

- For a single host, turn it on with **Log this session** (SSH, local shell) or **Log session to file** (Telnet, Serial) in the host editor.
- For all sessions: **Settings → Sessions → Log all sessions**.
- By default, logs go to the `logs` folder in the program data (see "Installation and updates"); choose a different folder with the **Browse…** button next to **Log folder**.
- Each session gets its own file, `<host name>_YYYY-MM-DD_HH-MM-SS.log`.

A log contains only text printed by the server, without control codes, and never the characters typed on the keyboard. Passwords typed in the terminal (e.g. for `sudo`) therefore never end up in a log. VNC sessions are not logged.

> **Note.** Logs are ordinary text files outside the encrypted vault. If the server prints confidential data (e.g. the contents of a configuration file with a password), it ends up in the log as plain text. Protect the log folder like other confidential documents.

24. Security

What is encrypted

All data saved in the program is stored in the vault, in the `rsessions.db` file. The content of every item is encrypted:

Encrypted	Plain in the vault file
hosts (names, addresses, settings), identities, passwords, private keys, sudo passwords and user passwords, snippets, tunnels, known hosts, connection history, settings, license	item identifiers, their kind (e.g. host, key), the links between them and save dates

- The vault key is derived from the password with Argon2id, a deliberately slow algorithm that makes password guessing harder. Data is encrypted with XChaCha20-Poly1305, which detects any change to the encrypted data.
- Every item has its own key, encrypted with the vault's master key. The master key is encrypted twice: with the password and with the recovery code.
- Deleted data is overwritten in the database file, not just marked as deleted.
- Passwords and keys are not shown unless needed; viewing a saved password requires entering the vault password again.

Outside the vault, in ordinary files, there are only: session logs (if enabled), working copies of files edited in an external program, and files you save yourself (e.g. downloaded from a server, or an exported key).

What the program does not send

R-Sessions has no user account and does not send usage statistics, server names or addresses, passwords, keys or license data. The program connects to the internet on its own in only two cases, each time with a plain request containing no data about you:

- the update check once a day (can be turned off in **Settings** → **About**);
- downloading the list of blocked licenses once a day after the vault is unlocked.

The **Get Pro** and **Renew** buttons open a page in the browser only when clicked; **Renew** passes only the license number in the address.

25. Troubleshooting

Problem	Cause	What to do
I forgot the vault password	the password cannot be recovered or reset	on the unlock screen click **Use recovery code** and set a new password (see "First start"); without the recovery code the data cannot be read

Problem	Cause	What to do
The host key has changed	server reinstall, key change or an interception attempt	confirm the change with the server's administrator and compare the fingerprint; only then **Replace key and connect**
The program keeps asking for a password, sign-in fails	wrong username or password, public key not added on the server	check the host's identity in **Keychain**; for a key, add the public key (**Copy public key**) to `~/.ssh/authorized_keys` on the server
The connection to a server drops when idle	a firewall or router closes idle connections	set **Keep-alive** in the host, e.g. 30 seconds
The statistics bar shows no data	the server does not run Linux or the option is off	select **Show server stats** in the host; statistics are not available on other systems
A tunnel shows **Failed**	the port on your computer is taken by another program, or the server refused the forward	choose a different **Listen port**; check that the SSH server allows port forwarding
A tunnel shows **Needs attention**	the tunnel is waiting for a host key to be accepted or a password	click the tunnel switch (**Connect**) and answer the program's prompt
A transfer shows "Waiting for connection"	the connection to the server was interrupted	the transfer resumes after reconnecting; check the network
No serial ports found	no driver for the USB adapter, or no permission for the port	install the driver; on Linux, add the user to the `dialout` group and sign in again
The serial port shows garbled characters or nothing	mismatched transmission parameters	check **Baud rate** and the other parameters in the device documentation
VNC: "The server needs a security type R-Sessions does not support yet"	the VNC server requires a sign-in method other than a VNC password	enable VNC password authentication on the VNC server; for encryption use **Connect through**
Free version limit reached	15 hosts or 2 tunnels are saved	load a Pro license, delete items you no longer need, or use Quick connect
"This license has been disabled..."	the license is on the blocked list	contact RALKOM
The update is not signed correctly and was not installed.	the downloaded update file is damaged or does not come from RALKOM	try again later or download the installer from the website; if the error persists, report it
No lock-on-sleep option on Linux	the system has no systemd-logind service	set **Auto-lock** and lock the vault manually (`Ctrl` + `Shift` + `L`)

26. Limitations of version 1.0

- No RDP (Windows Remote Desktop) connections. The protocol list contains an "RDP (not available yet)" item.
- The program runs on Windows and Linux; there is no macOS version.
- The user interface is available in English only.
- Files cannot be dragged between the panes and Windows Explorer or a Linux file manager; use the **Upload files…**, **Upload folder…** and **Download to…** buttons instead.
- The server statistics bar works only with Linux servers. **Follow terminal** requires a Linux server or a shell that reports the current folder.
- VNC supports servers with no password and with a VNC password. Other VNC server sign-in methods are not supported.
- Quick connect supports SSH only, without jump hosts or tunnels.
- The system tray icon is available on Windows only; on Linux, closing the window minimizes it.
- Bell notifications are not kept after the program closes.
- Each computer has its own vault; there is no synchronization between computers. Move the configuration with export.

27. Technical support

If you have a problem, contact RALKOM: `kontakt@ralkom.pl`, `www.ralkom.pl`. Please include:

- the program version and build number (**Settings** → **About**, e.g. "R-Sessions 1.0.0, Commit aed2ab5") and your system (Windows, or the Linux distribution and its version);
- the text of the message from the program window or the bell, and the time it appeared;
- for license problems: the license number from **Settings** → **License**.

Do not send your vault password, recovery code, private keys or the vault file. RALKOM will never ask for them.